Victor Sandoval

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RESEARCH OBJECTIVE

To pursue graduate research in cybersecurity, artificial intelligence, and technology-enhanced learning, with a focus on how emerging technologies can improve student engagement, technical communication, and technical education.

EDUCATION

The Pennsylvania State University - Penn State Greater Allegheny
Bachelor of Science in Information Technology | Minor in Security and Risk Analysis
Magna Cum Laude | GPA: 3.87/4.00 | May 2026

- **Relevant coursework:** Network Security; Computer and Cyber Forensics; Cyber Defense Studio; Legal and Regulatory Environment of Privacy and Security; Information Security Management

HONORS & AWARDS

- Student Marshal, Information Technology, Penn State Greater Allegheny, May 2026 - selected to represent the major at commencement after earning the highest GPA within the major.
- Honors Program Participant, Penn State Greater Allegheny, 2025-2026 Academic Year - recognized for two semesters of participation reflecting a strong commitment to academic excellence and scholarly engagement.
- First Place, Social and Behavioral Sciences - 2026 Penn State Undergraduate Exhibition for Research, Inquiry, and Creative Activity, University Park, PA; awarded for "Enhancing Information Technology and Cybersecurity Student Communication Skills through Immersive AR/VR and AI-Driven Simulations Using VirtualSpeech."
- First Place, Audience Choice Award - Fall 2025 Student Research and Creativity Conference, Penn State Greater Allegheny.
- Best Poster Presentation - Fall 2025 Student Research and Creativity Conference, Penn State Greater Allegheny.
- Dean's List, The Pennsylvania State University, Fall 2024 - Spring 2026.

RESEARCH EXPERIENCE

Information Technology & Cybersecurity Research, Penn State Greater Allegheny *McKeesport, PA*
Lead Undergraduate Research Assistant Sep 2025 - May 2026

- Led onboarding and technical training for 4 new undergraduate researchers, introducing research procedures, project workflows, documentation standards, and tools used across active faculty-led projects.
- Coordinated activities across 2 faculty-led research initiatives involving AR/VR, AI-driven simulations, cybersecurity education, and technology-enhanced learning, tracking weekly progress, decisions, and follow-up responsibilities.
- Presented ongoing research projects and undergraduate research opportunities to students at 4 area high schools, explaining project goals, technologies, and pathways for becoming involved in research.
- Maintained weekly research documentation across 2 semesters, tracking project decisions, assigned tasks, technical updates, and follow-up items for the undergraduate research team.

Information Technology & Cybersecurity Research, Penn State Greater Allegheny *McKeesport, PA*
Undergraduate Research Assistant Sep 2024 - Sep 2025

- Served as lead researcher on a Raspberry Pi study involving 15 undergraduate students, helping design and deliver hands-on activities using Raspberry Pi 5 devices, GPIO circuits, and Python programming to examine student motivation and engagement.
- Managed the VirtualSpeech platform for an immersive communication study involving 50 IT and Cybersecurity students across three courses, uploading research materials and assisting participants during AI-driven AR/VR simulations.
- Helped administer and support 2 distinct technology-enhanced learning studies, one involving Raspberry Pi/Python instruction and another involving AI-driven AR/VR communication training.

CONFERENCE PRESENTATIONS

Sandoval, V. (2026, April). *Enhancing Information Technology and Cybersecurity Student Communication Skills through Immersive AR/VR and AI-Driven Simulations Using VirtualSpeech.* Undergraduate Exhibition for Research, Inquiry, and Creative Activity, Penn State University Park, University Park, PA. Poster presentation; First Place, Social and Behavioral Sciences.

Moranelli, J., **Sandoval, V.**, Fullard, C., & Shaw, A. (2026, April). *Enhancing Information Technology and Cybersecurity Student Communication Skills through Immersive AR/VR and AI-Driven Simulations Using VirtualSpeech.* National Conference on Undergraduate Research (NCUR), Richmond, VA. Poster presentation.

Moranelli, J., **Sandoval, V.**, Fullard, C., & Shaw, A. (2025). *Enhancing Information Technology and Cybersecurity Student Communication Skills Through Immersive AR/VR and AI-Driven Simulations Using VirtualSpeech.* Fall 2025 Student Research and Creativity Conference, Penn State Greater Allegheny, McKeesport, PA. Poster presentation; First Place Audience Choice Award and Best Poster Presentation.

Breese, J. L., Cruz, L., Dupak, L., Lanka, S., **Sandoval, V.**, & Fullard, C. (2025, October). *Enhancing student motivation through a hands-on Raspberry Pi activity in an introductory IST course.* 65th International Association for Computer Information Systems Annual Conference, Clearwater, FL. Conference presentation; Sandoval served as the only undergraduate presenter.

Sandoval, V. & Neopaney, A. (2025, April). *Raspberry Pi GPIO Functionality through Interactive Learning.* National Conference on Undergraduate Research (NCUR), Pittsburgh, PA. Poster presentation.

PUBLICATION(S)

Breese, J. L., Cruz, L., Dupak, L., Lanka, S., **Sandoval, V.**, & Fullard, C. (2025). *Enhancing student motivation through a hands-on Raspberry Pi activity in an introductory IST course.* Issues in Information Systems, 26(4), 143-154. https://doi.org/10.48009/4_iis_2025_113

PROFESSIONAL EXPERIENCE

PNC Bank (Contractor via TEKsystems) *Pittsburgh, PA*
Platform Engineer May 2026 - Present

- Supported the development of 2 new data centers by creating and modifying more than 10 Ansible playbooks and YAML-based configurations.
- Contributed to the development of a GitHub Actions, AAP, and Nautobot-based network automation pipeline, researching and integrating 6 enterprise platforms including GitHub, AAP, CyberArk, JFrog, ServiceNow, and Confluence.
- Configured and tested automation workflows across 7 Cisco Nexus test devices (7K and 9K platforms), validating software upgrade, maintenance-mode, and credential management processes.
- Onboarded 2 production intent dataset (DCC Intent) into GitHub and supported the design of a Pull Request-driven workflow for automated network upgrades. Support the transition of development workflows from Bitbucket to GitHub by adapting repositories, collaboration practices, and version-control processes.
- Leveraged GitHub Copilot to support development, troubleshooting, and documentation of automation workflows and Ansible/YAML configurations, reviewing and refining AI-generated recommendations before implementation.

Rizzo International *Pittsburgh, PA*
Help Desk Technician Sep 2025 - May 2026

- Assisted with a network inventory system migration to Snipe-IT by auditing existing assets and migrating more than 50 network devices to improve asset tracking accuracy and visibility.
- Provided first-level IT support for approximately 115 employees, diagnosing and resolving 5+ hardware, software, and network issues per week to maintain day-to-day business operations.
- Managed Active Directory and Azure Admin Center access for approximately 115 users, creating and updating 100+ accounts and supporting permissions, access requests, and cloud-based administrative tasks.
- Configured and deployed 25+ employee workstations, printers, and peripheral devices, supporting new-user onboarding, hardware upgrades, and day-to-day office technology needs.

Westinghouse Electric Company *Pittsburgh, PA*
ISD Global Support Intern May 2025 - Aug 2025
- Performed daily infrastructure status checks across 18 Microsoft Teams Rooms, 5 TVs, and 4 data closets, helping maintain reliable functionality across collaboration spaces and supporting infrastructure.
- Set up and configured 2 Microsoft Teams Rooms by connecting mini-PCs, scheduling tablets, and VLAN-enabled ports.
- Upgraded 10 Microsoft Surface tablets from Windows 10 to Windows 11, configured domain settings, and installed required software.

Pittsburgh Hispanic Development Corporation *Pittsburgh, PA*
Information Technology Intern Jan 2025 - May 2025
- Designed and analyzed an organization-wide IT security assessment for approximately 30 employees, identifying gaps in password practices, phishing awareness, and device security.
- Delivered security recommendations to strengthen organizational practices and improve employee security awareness.

Achieva *Pittsburgh, PA*
Information Technology Intern Oct 2024 - Jan 2025
- Supported a phone-system conversion across 45 locations by deploying new 5G modems and documenting the IT environment for future maintenance and audits.
- Provided troubleshooting, repairs, upgrades, and general endpoint support for approximately 250 devices, resolving 50+ hardware and software issues to maintain reliable day-to-day operations.

Community College of Allegheny County - Enrollment Services *Pittsburgh, PA*
Information Technology Intern Jun 2024 - Aug 2024
- Managed and organized 2 SharePoint sites, updating content and maintaining site structure to support information sharing and collaboration.
- Assisted with a new network control room by installing Proxmox on 8 servers and configuring a virtual environment for research and classroom use.

LEADERSHIP

Cybersecurity Club, Penn State Greater Allegheny *McKeesport, PA*
Founder | President Dec 2024 - Jan 2026
- Set up a hacking seminar alongside an IT professor and a solutions engineer, educating approximately 10 students on ethical hacking, penetration testing, and cybersecurity best practices.
- Coordinated a hacking seminar with an IT professor and solutions engineer and arranged a Carnegie Mellon University guest speaker to introduce 15+ students to picoCTF.

Student Government, Penn State Greater Allegheny *McKeesport, PA*
Senator | Secretary Sep 2024 - May 2026
- Elected to the Student Government Executive Board as Secretary, supporting executive operations, organizational communication, attendance, and official meeting documentation.
- Represented a campus of over 300 students as a Senator and contributed to campus initiatives and student advocacy.

Lion Ambassador, Penn State Greater Allegheny *McKeesport, PA*
Tour Guide Sep 2024 - Sep 2025
- Guided over 50 prospective students and families through campus visits, sharing information about academics, student involvement, and opportunities at Penn State.

CERTIFICATIONS

CITI Program: Students in Research; Human Subjects Research

SKILLS

Languages/Scripting: Python, SQL, PowerShell, Ansible, Terraform, Git
Technical Skills: Networking, Troubleshooting, Software Deployment, Asset & Inventory Management, System Monitoring, Endpoint Configuration, IT Security Awareness, Risk Assessment, Technical Documentation
System/Platforms: Windows 10/11, Active Directory, Azure Administration, Microsoft 365, Amazon Web Services
Additional Skills: Business Writing, Technical Writing, Time Management, Creativity, Communication, Spanish